

1. The element **Me**, which due to the colour variety of its compounds is named after the nordic goddess of beauty, belongs to the d-elements.

The technical synthesis of **Me** starts from one of its oxides **A**. The corresponding ores undergo oxidation in a roasting process then are treated with alkaline salts and finally **Me** is isolated by leaching out the element.

The oxide **A** contains 43.98 % (m/m) of oxygen. In this compound the element **Me** shows an odd oxidation number. **Me** will be generated by reacting oxide **A** with metallic Ca at 950°C the element.

1.1 Which element is it? Support your reasoning by a calculation!

1.2 Write down the valence electron configuration of the element in its ground state.

1.3 Write down the formula and the name of oxide **A**?

1.4 Write a balanced equation for the reaction of the oxide with Ca.

In the following refining process, the element is inserted into molten NaCl as an electrode (the liquid contains **Me**³⁺-ions of the element). Using a voltage of 60 V, the pure element is precipitated at a tantal electrode. Before the electrolysis the mass of an electrode of this element is 500 g. For 16.0 hours an electrolysis takes place with a constant current of 50.0 A.

1.5 State which metals play the role of the cathode and the anode in the electrolytical refining process.

1.6 Calculate the mass of the element precipitated using the data from above, if the current yield is 90.0%.

1.7 Calculate the purity (in %) of the crude element **Me** used in the electrolysis.

The element **Me** forms a body centred lattice. In an X-ray structure analysis, an X-ray beam with the wave length $\lambda = 71.1$ pm is diffracted at the crystal plane (111). The diffraction angle for the first order beam has a value of 11.7°.

1.8 Calculate the lattice constant a_0 of the unit cell!

1.9 Calculate the atomic radius of the element **Me**.

1.10 Calculate the density of the element **Me**.

Treating the amphoteric oxide **A** with concentrated sodium hydroxide, a salt **B**, analogous to an orthophosphate, crystallizes as a decahydrate.

Adding an acid to very diluted solutions of this salt, the anion will be protonated to give the conjugated acid. Adding another proton will result in an elimination of two water molecules. Additionally, a faintly yellow coloured cation **C** is formed.

1.11 Write down the formula of salt **B**.

1.12 What is the formula of cation **C**?

The cation **C** may be reduced step by step. Taking one electron the blue cation **D** is generated, then the green cation **E** and finally cation **F**. In each of the first two steps an oxygen atom is split off. The standard redox potentials of all the steps mentioned and also the potential of the reduction to **Me** are (in acidic solution):

$$E^\circ(\text{C/D}) = 1.00 \text{ V}$$

$$E^\circ(\text{D/E}) = 0.359 \text{ V}$$

$$E^\circ(\text{E/F}) = -0.256 \text{ V}$$

$$E^\circ(\text{F/Me}) = -1.186 \text{ V}$$

- 1.13 Write down the formulae of the cations **D**, **E** and **F**.
- 1.14 Write down an equation for the half reaction of the redox pair **C/D**.
- 1.15 Draw a Frost-diagram of all the species of element **Me** mentioned above.
- 1.16 Which of these is the most stable one?
- 1.17 How many of these species are not stable against disproportionation?

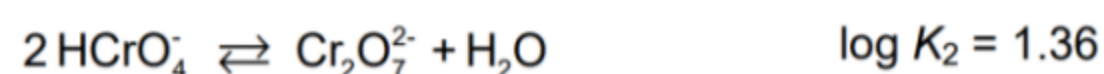
Element **Me** forms numerous coordination compounds with different coordination numbers and different oxidation states. Examples are:

- 1) $[\text{Me}(\text{bipy})_3]^+$
- 2) $[\text{Me}(\text{CO})_6]^-$
- 3) $[\text{MeO}(\text{O}_2)_2\text{ox}]^{3-}$
- 4) $[\text{MeOCl}_2(\text{N}(\text{CH}_3)_3)_2]$

- 1.18 Draw the MO-scheme of the two-dentate peroxo-anion!
- 1.19 State the bond order and the magnetic property of the peroxo ligand.
- 1.20 Give the systematic names of the complexes with the numbers **2** and **3**.
- 1.21 Determine the oxidation number of the centre in each of the complexes. Also state the geometry following VSEPR, and determine, whether the complex exhibits chirality.

2. The precipitation is widely used in classical methods of the quantitative and qualitative analysis for the separation of ions. The possibility of separation is determined by the equilibrium concentrations of all species in a solution to be analyzed.

Potassium dichromate is one of the most widely used precipitating reagents. The following equilibria are established in aqueous solutions of Cr(VI).



2.1 Calculate the equilibrium constants

- a) $\text{CrO}_4^{2-} + \text{H}_2\text{O} \rightleftharpoons \text{HCrO}_4^- + \text{OH}^-$
 b) $\text{Cr}_2\text{O}_7^{2-} + 2 \text{OH}^- \rightleftharpoons 2 \text{CrO}_4^{2-} + \text{H}_2\text{O}$

2.2 In what direction shall the equilibrium state 1b shift upon the addition of the following reagents to the aqueous solution of potassium dichromate?

- a) KOH b) HCl c) BaCl₂ d) H₂O

The solubility product of BaCrO₄ is 1.2×10^{-10} . BaCr₂O₇ is well soluble in water.

2.3 Calculate the pH value of the following solutions

- a) 0.010 M K₂CrO₄
 b) 0.010 M K₂Cr₂O₇
 c) 0.010 M K₂Cr₂O₇ + 0.100 M CH₃COOH

Dissociation constant of acetic acid $K_a = 1.8 \times 10^{-5}$.

2.4 Calculate the equilibrium concentrations of the following ions in the solution of 0.010 M K₂Cr₂O₇ + 0.100 M CH₃COOH

- a) CrO₄²⁻ b) Cr₂O₇²⁻

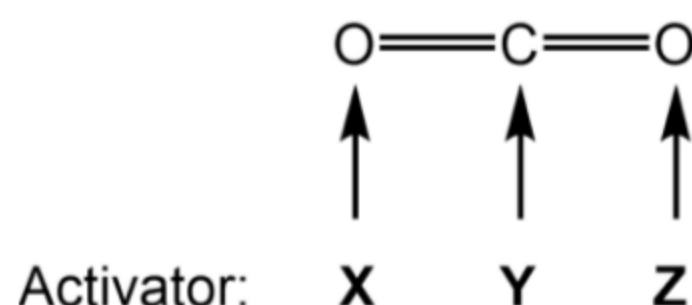
3. Many-faceted Oxides of Carbon

The trapping and conversion of CO₂ has become a major challenge in modern society. Whereas nature converts CO₂ elegantly into carbohydrates via the Calvin cycle with the help of solar energy, industry uses the Sabatier reaction to convert CO₂ to methane. Along this path, partially reduced products such as formic acid represent important intermediates. Conversion of CO₂ requires its activation, and modern research in homo- and heterogeneous catalysis for this reaction is steadily growing. Essentially any conversion of CO₂ requires H₂, as a co-reactant. Thus, the activation of CO₂ also requires the activation of H₂ at the same time. This leads to very interesting chemistry that includes catalyst design, mechanistic studies and biomimetic chemistry.

3.1 Complete the following reaction sequence with 2 H⁺ and 2 e⁻ in each step and the eventual side products. Draw the structures of A and B.



3.2 To activate CO₂, it must be polarized through non-covalent interactions. Describe the chemical properties associated with polarity of potential activators as indicated in the following scheme.



CO₂ and ammonia (NH₃) react to form urea (CO(NH₂)₂), which is a widely used fertilizer in agriculture. Before NH₃ was available in large quantities, CaCO₃ was reacted with carbon to get calcium carbide (acetylide, CaC₂). CaC₂ reacts at high temperatures with N₂ to give CaCN₂. CaCN₂ was used as a fertilizer since it forms urea with water in the soil.

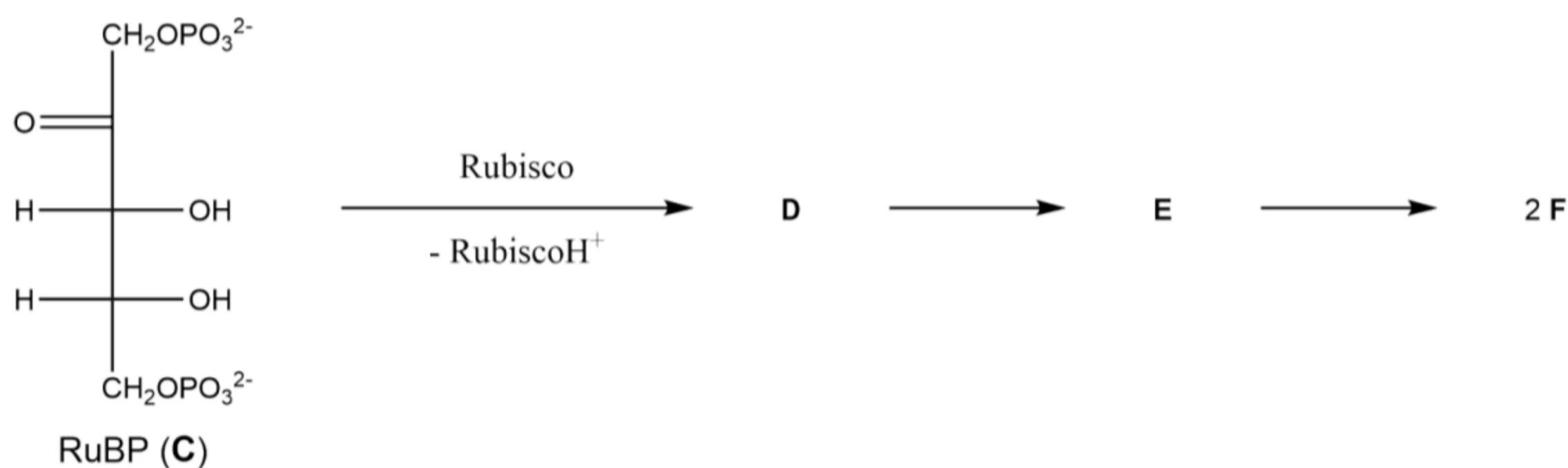
3.3 Draw the full reaction sequence from CaCO₃ to urea.

3.4 Cyanamid H₂CN₂ exists in two tautomeric forms. Draw the Lewis structures of both forms.

3.5 Sequestering CO₂ from the atmosphere is commercially done with different trapping agents. Draw the products obtained from reacting CO₂ with the following compounds.

1. primary amines R-NH₂
2. 1,2-diamines
3. glycerol
4. butadiene (catalyzed)

Starting from ribulose 1,5-bisphosphate RuBP (compound C), by action of the enzyme rubisco, a proton at C3 of RuBP is abstracted to form an enediolate D, which then in a nucleophilic attack binds CO₂. The resulting β-ketoacid E is immediately attacked by a water molecule at C3 to form a labile adduct that is split into two molecules of 3-phosphoglycerate F.



3.6 Draw the structures D, E and F for this reaction sequence

Besides CO₂, there are a number of other neutral binary oxides of carbon, namely CO₃, C₃O₂, and C₁₂O₉. The latter is the anhydride of a hexacarboxylic acid.

3.7 Draw the structures of these binary carbon oxides.

1. CO_3
2. C_3O_2
3. C_{12}O_9

Carbon monoxide CO binds very well to low valent transition metals. The highest stability in such complexes is achieved when all available orbitals are filled.

3.8 Draw a complex with elementary Ni^0 and CO. Draw its structure according to VSEPR.

3.9 Draw a complex with Fe^0 instead of Ni^0 . Draw its structure according to VSEPR.

Squaric acid is an uncommon acid, which liberates 4 equivalents of CO after a two-electron oxidation.

3.10 Draw the structure of squaric acid.

3.11 Squaric acid is a rather strong organic acid ($\text{pK}_{a1} = 1.5$, $\text{pK}_{a2} = 3.4$). Explain this observation.

3.12 Propose a structure of a different acid that upon oxidation would liberate 5 equivalents of CO

4. Until 1968, it was believed that anion **A1** and its salts were unstable. However, in 1968, Evan H. Appelman performed an unusual synthesis of **A1**. A 90% enriched isotope of element **X** was first irradiated with thermal neutrons, then dissolved in nitric acid and oxidised by ozone in alkaline solution, giving anion **B**, which within a few minutes spontaneously turned into **A1** without absorbing or releasing any chemical substances. 1.613 g of anion **B** was obtained from 1.000 g of **X**.

After the first unusual but successful attempt, scientists managed to oxidise anion **A2** to **A1** using electrolysis. For this purpose, Pt electrodes were immersed into 3.0 mL of the solution containing **A2** (solution **S0**) and into HClO_4 solution. Electrolysis was carried out with a current strength of 10 A/cm^2 for 1 hour. The work surface of each Pt electrode was 0.785 mm^2 , the current yield for **A1** was 11.5%.

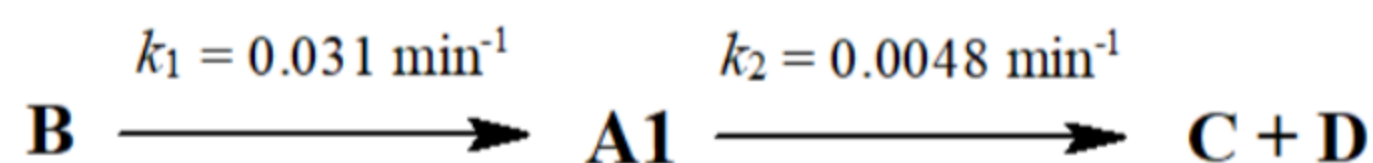
1. **Write** the equation of the half-reaction occurring at the cathode and **calculate** the total volume (in mL at 0°C , 1 atm) of the gas released at the cathode during electrolysis.

After electrolysis, the analyte contained a mixture of anions **A1** and **A2** (colourless solution **S1**). The Pt electrode was removed from the analyte, an excess of diluted ($\sim 1 \text{ M}$) acidic solution containing anion **A3** was added, and a yellow solution **S2** (containing elemental substance **A4**) was obtained. **A4** was removed by bubbling argon through **S2**, and a colourless solution **S3** was obtained. An excess of saturated ($\sim 10 \text{ M}$) acidic solution containing **A3** was added to solution **S3**, and a yellow solution **S4** (also containing elemental substance **A4**) was obtained. The whole volume of **S4** was diluted to 100.0 mL, and thus solution **S5** was

obtained. An excess of acidified KI solution was added to a 10.0 mL aliquot of solution **S5**, and the I_2 formed was titrated with 0.100 M $Na_2S_2O_3$ solution. A titre of 1.35 mL of $Na_2S_2O_3$ solution was obtained. Another 3.0 mL sample of solution **S0** was not subjected to electrolysis. Instead, it was diluted to 500.0 mL giving solution **S6**. An excess of acidified KI solution was added to a 10.0 mL aliquot of **S6**, and the I_2 formed was titrated with 0.100 M $Na_2S_2O_3$ solution. A titre of 10.10 mL of $Na_2S_2O_3$ solution was obtained.

2. **Write** the equation of the reaction occurring during titration between I_2 and $Na_2S_2O_3$.
3. **Determine** the chemical composition of **A1–A4**, **B**, and **X** (with atomic mass of enriched isotope).
Note: A1–A4 contain the same element.
4. **Calculate** the molar concentration (in M) of **A2** in solution **S0**.

A1, formed within a few minutes from anion **B**, spontaneously decomposes within a few hours into gaseous elemental substances **C** and **D** in a molar ratio 1:2. The reaction scheme shows the formation and decay of **A1** with the corresponding rate constants



5. **Determine** the detailed chemical composition of **C** and **D**. Both reactions shown in the scheme above are first order with respect to the corresponding reactants.
6. **Calculate** the time (t , in min) after the start of the decomposition of **B**, which corresponds to the maximum concentration of **A1**.

It was subsequently discovered that XeF_2 also oxidises **A2** to **A1**. Interestingly, another binary compound, **E**, of xenon (which does not contain fluoride) was also obtained by exactly the same unusual method (like $\mathbf{B} \rightarrow \mathbf{A1}$) from a potassium-containing salt **F** with a mass fraction of potassium of 12.62%.

7. **Determine** the detailed chemical composition of compounds **E** and **F**.